

**Performance-Based
Statement of Work (SOW)
Performance and Reliability Testing of the
IMPULSE CARTRIDGE and OUTPUT INITIATOR for the B-1B SEAT RETRACTION THRUSTER
DODICs MT92 and MT10
P/Ns 816951 and 816952
NSNs 1377-01-279-7180 and 1377-01-279-7181**

1.0 General

1.1 Introduction

The Naval Surface Warfare Center Indian Head Division (NSWC IHD) specializes in life cycle support for cartridge-and propellant-actuated devices (CAD/PADs). This includes exploratory development, qualification, acquisition, and comprehensive product and logistics support. The CAD/PAD Division administers, manages, and conducts development, engineering, procurement, and logistics support functions for CADs and PADs and Aircrew Escape Propulsion Systems (AEPS) for the DoD.

1.2 Scope

NSWC IHD requires test support in order to evaluate the performance and reliability level of the items listed in Table I. This requires performing non-destructive tests and ballistic/functional tests in accordance with the document(s) listed in Section 2.0 and providing data sheets in addition to testing summaries to the Government. The provided results will be compared directly with lot acceptance test (LAT) data to ascertain the effects of age and service exposure on performance and reliability. Quantities listed are approximate and could change, however; the total quantity shall not exceed what is listed in Table I. Additionally, the government reserves the right to test less than the number listed in Table I if required.

Table I. Test Sample

DODIC	NSN(s)	Part Number(s)	Quantity
MT92, MT10	1377-01-279-7180 and 1377-01-279-7181	816951 and 816952	45 of each P/N (90 total assets)

Throughout this document, the items listed in Table I will be referred to as the 'test sample'. For the purposes of this document, no interpretation shall be made for the phrase 'test sample' other than to refer to the items listed in Table I.

2.0 Applicable Documents

All testing shall be completed in accordance with:

Document	Title
ATP 700132	Pacific Scientific Energetic Materials Company Acceptance Test Procedure for B-1B Seat Retraction Thruster

The most current revision of the ATP shall be used for testing unless otherwise directed by the government.

2.1 Requirements

The contractor shall conduct and document an ordnance assessment test and analysis for the test sample. All tests subsequently listed in this document shall be completed in accordance with the document(s) listed in Section 2.0.

In the event that there have been changes to any of the documents in Section 2.0, the government shall be notified of the change and provided a copy for concurrence. The contractor shall provide the instrumentation, test equipment, gauges or any other equipment needed to support this effort.

The testing shall be completed within one-hundred and twenty (120) days after receipt of the government furnished material (GFM) test sample, including the delivery of data. The contractor shall provide a report of all non-destructive and destructive test results in accordance with Sections 2.2 through 2.3, and as outlined in Section 3.0.

2.2 Non-Destructive Evaluation

All non-destructive testing shall be conducted on the test sample in accordance with the applicable documents listed in Section 2.0. This shall be completed to provide a record of the pre-fire condition of the test sample in the event of a performance anomaly or failure during performance testing.

All NDT results shall be provided to the government, including a copy of the x-rays (radiographic film or digital). NDT shall be completed in the order listed in section 2.2.

2.2.1 Visual Inspection

A visual inspection of the test sample shall be completed to identify any physical anomalies. This includes identifying any damage that could affect function or installation. Any damage identified shall be documented and photographed. Inspection data shall include a record of the lot number, serial number, and date of manufacture.

2.2.2 Radiographic Inspection

A radiographic inspection of the test sample shall be completed in accordance with the applicable documents in Section 2.0 to identify any internal anomalies. A copy of the x-rays (radiographic film or digital) shall be provided to the government, along with the radiographic interpretation report. The government shall be notified of any anomalies or defects identified during radiographic inspection prior to ballistic/functional testing.

2.2.3 Leakage Inspection

A leakage inspection of the test sample shall be completed in accordance with the applicable documents in Section 2.0 to identify any internal anomalies. The government shall be notified of any anomalies or defects identified during the inspection prior to ballistic/functional testing.

2.3 Ballistic/Functional Evaluation

The test sample shall be ballistically/functionally tested in accordance with the applicable documents listed in Section 2.0 to verify that all applicable specification requirements are met.

The contractor shall also provide a summary of the ballistic test results (data sheets) after data reduction, also listing the acceptance criteria.

2.3.1 Performance Testing

Prior to performance testing, the test sample shall have completed the NDT listed in Section 2.2 in accordance with the applicable documents in Section 2.0.

Additionally, the test sample shall be temperature conditioned: (1) as specified in the government-provided sample identification/temperature breakdown, or (2) government concurrence has been provided for a contractor-provided sample identification/temperature breakdown. Government concurrence is required before temperature conditioning the test sample.

Performance firing for the test sample shall be completed within the specified timeframe of removal from the conditioning chamber in accordance with the applicable documents in Section 2.0.

2.3.2 Performance Testing Failures

In the event of a test failure, the contractor shall verify the test setup and complete a post-fire x-ray of the failed unit(s) alongside a successfully functioned unit. If possible, an unfired unit should also be included in the x-ray. The contractor must retain any failed units for at least 90 days after testing or until released for disposal by a government representative, whichever occurs first. Contractor shall ship failed unit(s) back to the government upon request.

Unless otherwise stated in the ATP, or at the discretion of the government witness(es), testing shall continue after a failure is observed.

2.3.3 Other

The contractor shall be responsible for disposing of all remaining test samples and any related hardware upon completion of the test.

3.0 Deliverables

- a. Ballistic traces for each performance test, with one trace being clearly marked as to where the specification parameters are being read/measured/interpreted (if applicable)
- b. Summary of the ballistic results (data sheets) after data reduction that also lists acceptance criteria
- c. Non-destructive inspection results excluding x-ray reports and a copy of the x-rays (film or digital)
- d. Description of the item listed in Table I including components and a summary of how it functions
- e. Description of the test procedures

- f. Pictures and block diagram of the test setup with call outs
- g. Pictures and block diagram of the item listed in Table I with call outs.
- h. Inspection records of the associated test fixture and data acquisition equipment including details of the setup such as sample rates, filtering (in-line or post), smoothing/averaging, calibration, etc...
- i. Failure results as described in Section 2.3.2 (if applicable)
- j. LAT data for the lots represented in the test samples, to include the means and standard deviations per parameter and temperature (if applicable)

CDRL	Spec	Sections	Title
A001	DI-MISC-80508B	2.0 – 3.0	Data Sheets, Drawings, and Pictures

4.0 Test Witnessing

The NSWC IHD Ordnance Assessment Team shall be responsible for monitoring and witnessing all requirements imposed under this SOW to ensure compliance. The NSWC IHD point of contact (POC) shall be notified of all programmed testing at least 20 working days prior to commencement of the test. The NSWC IHD POC reserves the right to accept or reject any proposed test date by the vendor due to scheduling conflicts. THE NSWC IHD POC must approve any proposed test date.

5.0 Period of Performance (POP)

Period of performance: 120 days after receipt of GFM. This includes test data, which shall be delivered no later than 30 days after completion of testing.

6.0 Disclaimer Statement

All reports resulting from this contract shall contain the following disclaimer statement on the cover of such reports:

“The views, opinions, and findings contained in this report are those of the author(s) and should not be construed as an official Department of Defense position, policy, or decision, unless so designated by other official documentation.”

7.0 Data Delivery and Points of Contact (POC)

Data shall be provided and delivered after the completion of testing. Delivery shall be completed via email, DoD SAFE (or equivalent, DoD-approved-and-compliant files sharing), or standard mail to the NSWC IHD POC.

Digital files shall be provided in the form of Microsoft Office documents, .pdf, .jpg, or .png.

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